MIL-W-3448D
25 February 1980
SUPERSEDING
MIL-W-3448C
6 July 1973

MILITARY SPECIFICATION

WALLBOARD: PACKING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for the packing of composition wallboard, insulating wallboard, gypsum wallboard, gypsum sheathing board and gypsum plasterboard.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-N-105	- Nails, Brads, Staples and Spikes, Wire, Cut and Wrought.
NN-P-530	- Plywood, Flat Panel.

FSC PACK

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, Virginia 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-S-781	- Strapping, Steel, and Seals.
MMM-A-260	- Adhesive, Water-Resistant, (For Sealing Waterproofed Papper).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officers.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF COMMERCE

Product Standard

PS 1	- Softwood Plywood, Construction and Industrial.
PS 17	- Polyethylene Sheeting.
PS 51	- Hardwood and Decorative Plywood.
PS 58	- Basic Hardboard.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

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3. REQUIREMENTS

3.1 Preproduction (first article) pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to producing production packs, that his applied packing and marking comply with the requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.2 Materials. Materials shall be as specified herein. Materials shall be free of defects affecting serviceability (see 6.4).

3.3 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

3.3.1 Level A. Wallboard of like description shall be packed in accordance with figure 1 or figure 2, as follows: Sheets of wallboard shall be carefully and uniformly stacked with the finished surfaces of each two sheets placed together. Sheet sizes 3-1/2 foot by 4-foot shall be stacked side by side (using 4-foot as to the width dimension). Quantity of sheets per stack shall be as specified (see 6.2), but shall not exceed 3,000 pounds net weight. Each stack shall be wrapped with barrier material conforming to PPP-B-1055, class M-1, with all folds, seams, and laps sealed with adhesive conforming to MMM-A-260, grade B, type and class optional or each stack shall be wrapped with black polyethylene film conforming to PS 17, minimum 6 mil thickness, with all seams, folds, and laps sealed to prevent the entrance of moisture.

3.3.2 Level B. Wallboard of like description shall be packed in accordance with figure 3 or figure 4. Wallboard shall be stacked as specified in 3.3.1. Stacked wallboard shall be protected with a polyethylene shroud conforming to PS 17, minimum thickness 6 mil. The shroud shall be secured in place.

3.3.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

3.4 Marking. Military marking shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contactor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction (first article) pack inspection.

4.3.1 Examination. The pack shall be examined for the defects specified in table I. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a completed pack of each level, prepared for shipment, shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance wth 4.4.2 shall be examined for the defects listed in table I. AQL shall be 2.5 percent defective.

TABLE I - Examination schedule.

Number	Defect	Requirement Para.
101.	Materials not as specified for level A or B.	3.3
102.	Finished surfaces of sheets not placed together for level A or B.	3.3.1
103.	Weight of sheets per pack exceeds 3000 pounds for level A or B.	3.3.1
104.	Barrier material not as specified for level A or B.	3.3.1
105.	Barrier material not sealed as specified for level A.	3.3.1
106.	Strapping not as specified.	See figs.
107.	Marking illegible, incomplete, incorrect, or missing.	3.4

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5. PACKAGING
(Not applicable to this specification.)

6. NOTES

6.1 Intended use. This specification is intended to be used for reference in commodity specifications and for direct reference in contracts or orders for packing of wallboard.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.1).
(c) Degree of packing required (see 3.3).
(d) Quantity of sheets to be packed in a stack (see 3.3.1).

6.3 Preproduction (first article) pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to pack and mark the wallboard in accordance with this specification.

6.4 Recycled materials. It is encouraged that recycled materials be used, when practical, as long as it meets the requirements of this specifications (see 3.2).

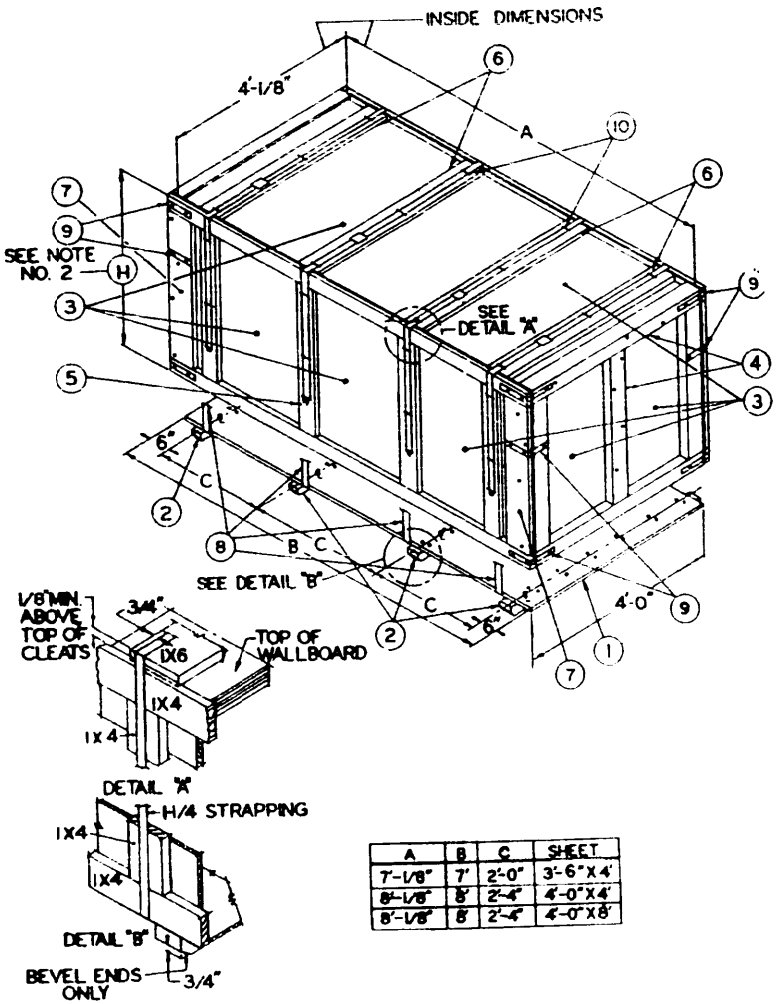
Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Army - SM

Preparing activity:
Army - ME

Project PACK 0602

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A	B	C	SHEET
7'-1/8"	7'	2'-0"	3'-6"X4'
8'-1/8"	8'	2'-4"	4'-0"X4'
8'-1/8"	8'	2'-4"	4'-0"X8'

FIND NO.	QTY	NOMENCLATURE
1	1	BASE PLYWOOD NN-P-530; GROUP A (PS 5), TYPE I, GRADE 3-4) OR GROUP B
		(PS 1, GRADE CD INTERIOR WITH EXTERIOR GLUE), 1/2" THICKNESS.
2	4	SKID LUMBER MIL-STD 731 CL2, MINIMUM 3"X4" NOMINAL SIZE (2"X4" PLUS 1"X4" RUBBING STRIP
		ACCEPTABLE). BEVEL ENDS 3/4" ON 45°.
3	5	ALL PLYWOOD PANELS (2 SIDES, 2 ENDS, 1 TOP) NN-P-530 GROUP A (PS 5), TYPE I,
		GRADE 3-4) OR GROUP B (PS 1, GRADE CD INTERIOR WITH EXTERIOR GLUE),
		1/4" THICKNESS. AT THE OPTION OF CONTRACTOR, PANELS MAY BE
		FABRICATED OF HARDBOARD OF 1/4" THICKNESS CONFORMING TO REQUIREMENTS OF
		PS 58, SERVICE TYPE OR BETTER, WITH A MINIMUM DENSITY OF 55 LBS/CU FT.
4	10	ALL CLEATS ON END PANELS TO BE NOMINAL 2"X4".
5	16	ALL CLEATS ON SIDE PANELS TO BE NOMINAL 1"X4" EXCEPT AS INDICATED.
6	4	ALL TOP FILLER STRIPS TO BE NOMINAL 1"X6".
7	4	ALL SIDE PANEL END CLEATS TO BE NOMINAL 1"X8".
8	4	STRAPPING QQ-S-781 TYPE I, FINISH B, .75"X.031" MINIMUM, TO BE LOCATED
		ADJACENT TO SKIDS.
9	12	CORNER STRAPPING QQ-S-781, TYPE II, FINISH A, .75"X.028"X13".
10	AR	STAPLES 1-3/8" CROWN WIDTH, 3/4" LONG GALVANIZED STEEL

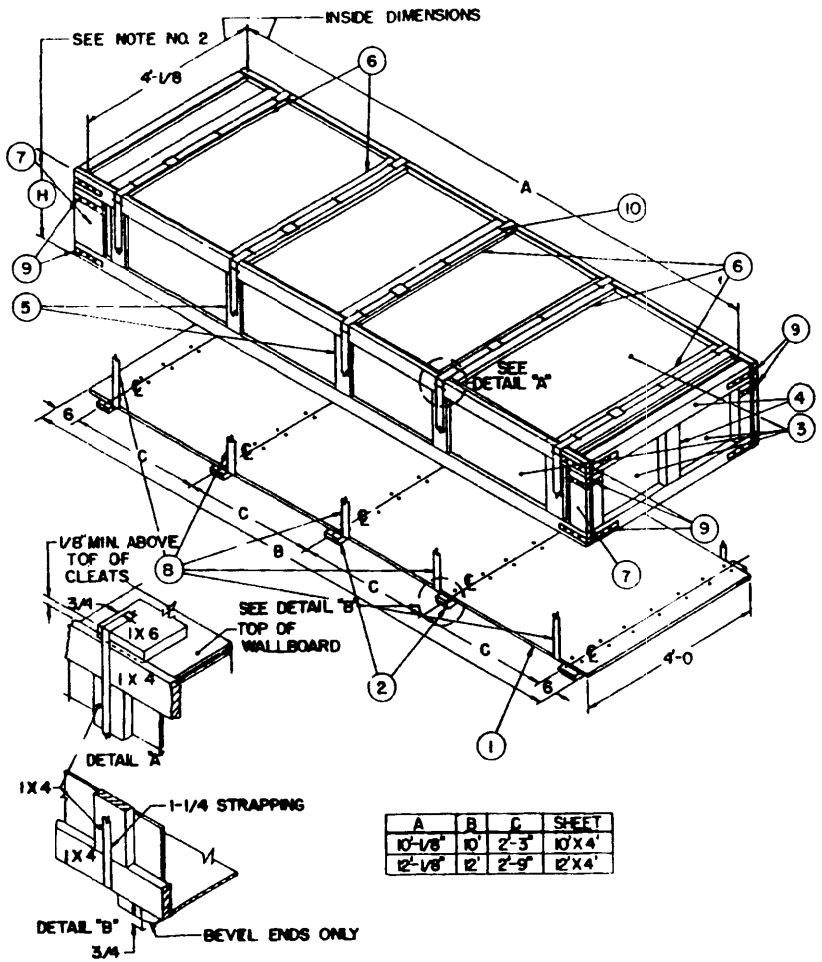
NOTES:

1. CONSTRUCTION OF PANELS TO BE IN ACCORDANCE WITH REQUIREMENTS OF PPP-B-601.
2. HEIGHT DEPENDENT UPON QUANTITY OF WALLBOARD SHEETS TO BE CONTAINED.
3. NET WEIGHT NOT TO EXCEED 3000 LBS.
4. PANELS TO BE CONSTRUCTED BY NAILING THROUGH PLYWOOD INTO CLEATS AND CLINCHING ON OUTSIDE OF CLEATS, WHERE APPLICABLE.
5. DIMENSIONS SHOWN ARE BASED ON NOMINAL LUMBER SIZES AS WELL AS ESTABLISHED COMMERCIAL STOCK SIZES OF WALLBOARD BEING PACKED.
6. HEIGHT OF SIDE PANELS TO BE SUCH THAT WHEN CLEATS (7) ARE UNDER COMPRESSION OF STRAPPING (8) THEY EXTEND NOT LESS THAN 1/8" ABOVE SIDE PANEL (SEE DETAIL "A").
7. NAILING OF 1"X8" SIDE PANEL END CLEATS TO END PANEL CLEATS: USE 12d CEMENT COATED SINKERS (FF-N-105, TYPE II, STYLE 8).
8. NAILING OF CLEATS: USE 7d CEMENT COATED COOLERS OR SINKERS (FF-N-105 TYPE II, STYLE 7 OR 8).
9. NAILING OF CORNER STRAPPING: USE 1" LONG ROOFING NAILS (FF-N-105 TYPE II, STYLE 20, STEEL WIRE, FLAT REINFORCED HEAD, NEEDLE OR DIAMOND POINT, ROUND SMOOTH SHANK, BRIGHT OR ZINC COATED.
10. NAILING OF BASE TO SKIDS: USE 10d CEMENT COATED COOLERS OR SINKERS (FF-N-105, TYPE II, STYLE 7 OR 8). IF SKIDS ARE MADE OF TWO-PIECE CONSTRUCTION, SECURE BEVELED RUBBING STRIP TO NOMINAL 2"X4" SKID WITH 8d CEMENT COATED COOLERS OR SINKERS (FF-N-105, TYPE II, STYLE 7 OR 8).

FIGURE 1. Packing level "A" (to 8' length).

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A	B	C	SHEET
10'-1/8"	10"	2'-3"	10'X4'
12'-1/8"	12"	2'-9"	12'X4'

FIND NO.	QTY	NOMENCLATURE
1	1	BASE PLYWOOD NN-P-530, GROUP A (PS 51, TYPE 1, GRADE 3-4) OR GROUP B
		(PS 1, GRADE CD INTERIOR WITH EXTERIOR GLUE), 1/2" THICKNESS.
2	5	SKID LUMBER MIL-STD 731 CL2, MINIMUM 3"X4" NOMINAL SIZE (2"X4" PLUS 1"X4"
		RUBBING STRIP ACCEPTABLE) BEVEL END 3/4" ON 45°.
3	5	ALL PLYWOOD PANELS (2 SIDES, 2 ENDS, 1 TOP) NN-P-530 GROUP A (PS 51,) TYPE I,
		GRADE 3-4) OR GROUP B (PS 1, GRADE CD INTERIOR WITH EXTERIOR GLUE),
		1/4" THICKNESS. AT THE OPTION OF CONTRACTOR, PANELS MAY
		BE HARDBOARD OF 1/4" THICKNESS CONFORMING TO REQUIREMENTS OF PS 58,
		SERVICE TYPE OR BETTER, WITH A MINIMUM DENSITY OF 55 LBS/CU. FT.
4	10	ALL CLEATS ON END PANELS TO BE NOMINAL 2"X4".
5	16	ALL CLEATS ON SIDE PANELS TO BE NOMINAL 1"X4" EXCEPT AS INDICATED.
6	5	ALL TOP FILLER STRIPS TO BE NOMINAL 1"X6".
7	4	ALL SIDE PANEL END CLEATS TO BE NOMINAL 1"X 8".
8	5	STRAPPING QQ-S-781 TYPE I, FINISH B, .75"X.031" MINIMUM, TO BE
		LOCATED ADJACENT TO SKIDS.
9	12	CORNER STRAPPING QQ-S-781, TYPE II, FINISH A, .75"X.028"X13".
10	AR	STAPLES 1-3/8" CROWN WIDTH, 3/4" LONG, GALVANIZED STEEL.

NOTES

1. CONSTRUCTION OF PANELS TO BE IN ACCORDANCE WITH REQUIREMENTS OF PPP-B-601.
2. HEIGHT DEPENDENT UPON QUANTITY OF WALLBOARD SHEETS TO BE CONTAINED.
3. SIDE PANELS MAY BE FABRICATED OF 8' PLYWOOD SHEET WITH FILLER STRIP ON EACH END TO EQUAL REQUIRED LENGTH.
4. NET WEIGHT NOT TO EXCEED 3000 LBS.
5. PANELS TO BE CONSTRUCTED BY NAILING THROUGH PLYWOOD INTO CLEATS AND CLINCHING ON OUTSIDE OF CLEATS, WHERE APPLICABLE.
6. DIMENSIONS SHOWN ARE BASED ON NOMINAL LUMBER SIZES AS WELL AS ESTABLISHED COMMERCIAL STOCK SIZES OF WALLBOARD BEING PACKED.
7. HEIGHT OF SIDE PANELS TO BE SUCH THAT WHEN CLEATS (7) ARE UNDER COMPRESSION OF STRAPPING (8) THEY EXTEND NOT LESS THAN 1/8" ABOVE SIDE PANEL (SEE DETAIL "A").
8. NAILING OF 1"X 8" SIDE PANEL END CLEATS TO END PANEL CLEATS. USE 12d CEMENT COATED SINKERS (FF-N-105, TYPE II, STYLE 8).
9. NAILING OF CLEATS: USE 7d CEMENT COATED COOLERS, OR SINKERS (FF-N-105, TYPE II, STYLE 7 OR 8).
10. NAILING OF CORNER STRAPPING: USE 1" LONG ROOFING NAILS (FF-N-105, TYPE II, STYLE 20, STEEL WIRE, FLAT REINFORCED HEAD, NEEDLE OR DIAMOND POINT, ROUND SMOOTH SHANK, BRIGHT OR ZINC COATED).
11. NAILING OF BASE TO SKIDS: USE 10d CEMENT COATED COOLERS OR SINKERS (FF-N-105, TYPE II, STYLE 7 OR 8). IF SKIDS ARE MADE OF TWO-PIECE CONSTRUCTION SECURE BEVELED RUBBING STRIP TO NOMINAL 2"X4" SKID WITH 8d CEMENTED COATED COOLERS OR SINKERS (FF-N-105, TYPE II, STYLE 7 OR 8).
12. PLYWOOD BASE TO BE CONSTRUCTED WITH MINIMUM NUMBER OF JOINTS. JOINTS TO BE MADE ON CENTER-LINE OF SKID.

FIGURE 2. Packing level "A" (over 8' length).

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FIND NO.	QTY	NOMENCLATURE
1	6	1/2" OR GREATER WALLBOARD SHEATHING
2	3	SKIDS 3"X4" NOM.(2"X4" PLUS 1"X4" RUBBING STRIP ACCEPTABLE). ENDS BEVELED
3	17	NOM. 1"X6" LUMBER STRIP
4	5	STRAPPING QQ-S-781 TYPE I OR II FINISH A 3/4"X.031" LENGTH AR.
5	AR	STAPLES

NOTES:
1. HEIGHT DETERMINED BY QUANTITY OF WALLBOARD TO BE CONTAINED
2. NET WEIGHT NOT TO EXCEED 3000 LBS.
3. NAILING OF LENGTHWISE STRIPS TO CROSS SKIDS: CEMENT COATED COOLERS OR SINKERS OF SUFFICIENT LENGTH TO ALLOW CLINCH OF MIN. 1/8 INCH.

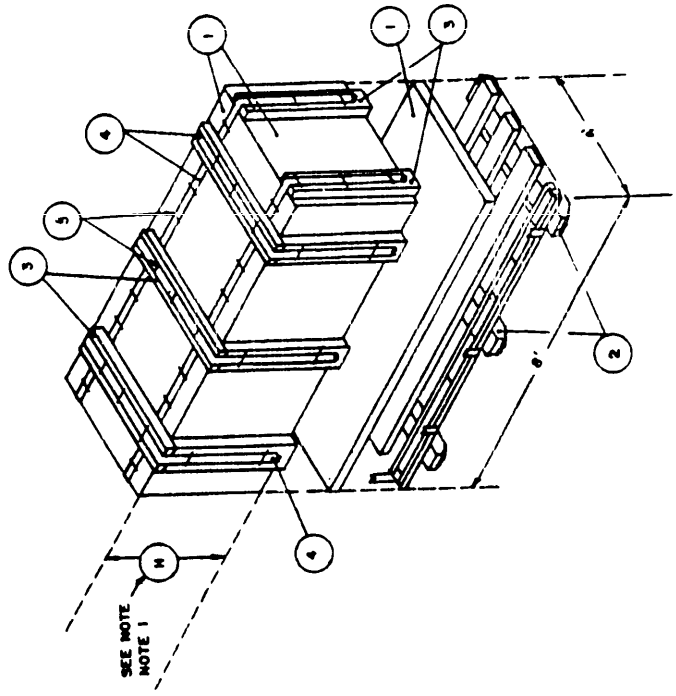
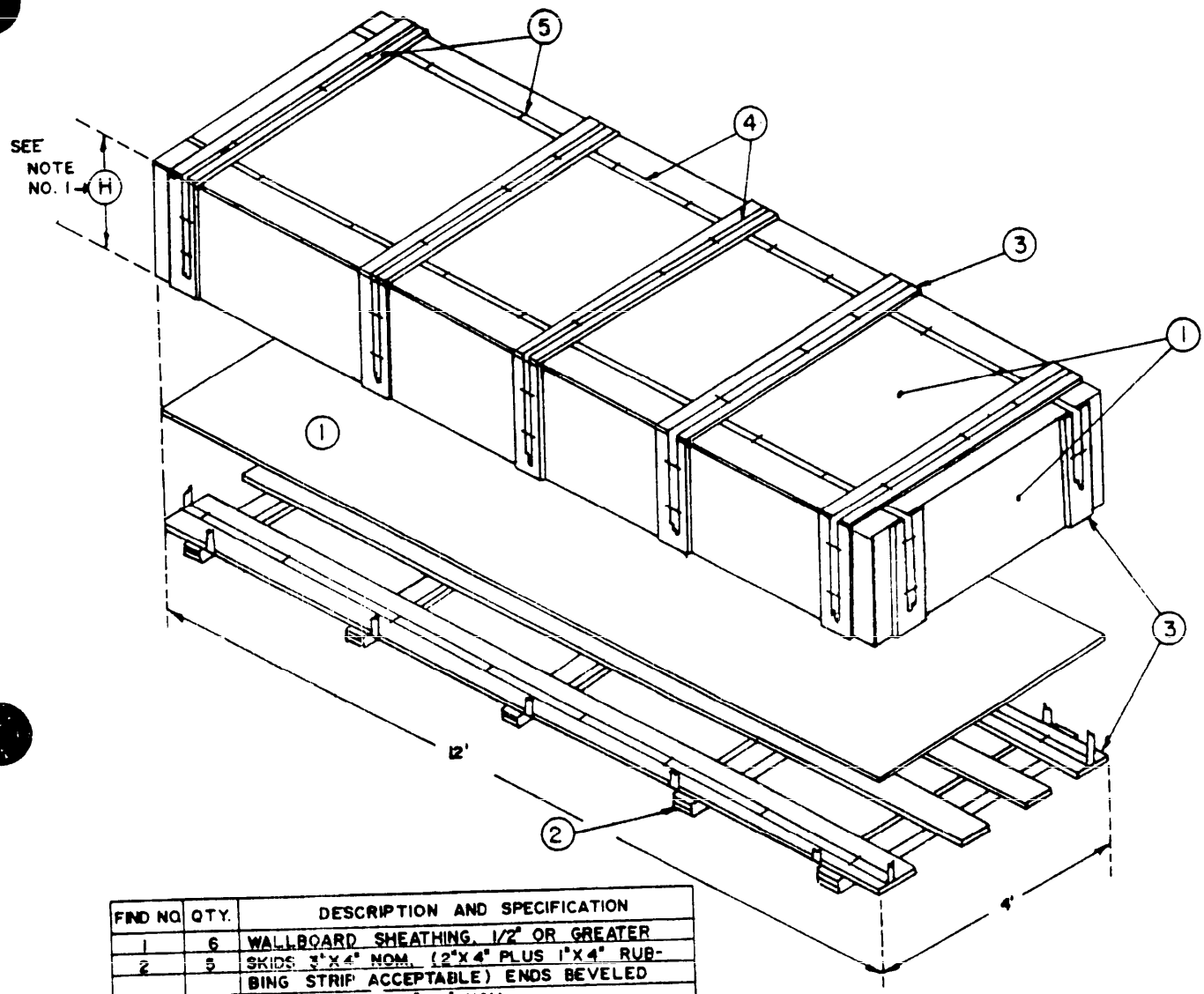


FIGURE 3. Packing level "B" (to 8' length).

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FIND NO	QTY.	DESCRIPTION AND SPECIFICATION
1	6	WALLBOARD SHEATHING, 1/2" OR GREATER
2	5	SKIDS: 3"X4" NOM. (2"X4" PLUS 1"X4" RUB-BING STRIP ACCEPTABLE) ENDS BEVELED
3	23	LUMBER STRIP, 1"X6" NOM.
4	7	STRAPPING, QQ-S-781, TYPE I OR IV, FINISH A, 3/4"X.031" LENGTH AR
5	AR	STAPLES

NOTES:

1. HEIGHT DETERMINED BY QUANTITY OF WALLBOARD TO BE CONTAINED.
2. NET WEIGHT NOT TO EXCEED 3000 LBS.
3. NAILING OF LENGTHWISE STRIPS TO CROSS SKIDS; CEMENT COATED COOLER OR SINKERS OF SUFFICIENT LENGTH TO ALLOW MINIMUM CLINCH OF 1/8-INCH.

FIGURE 4. Packing level "B" (over 8' length).

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DOCUMENT IDENTIFIER AND TITLE

MIL-W-3448D Wallboard: Packing of

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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